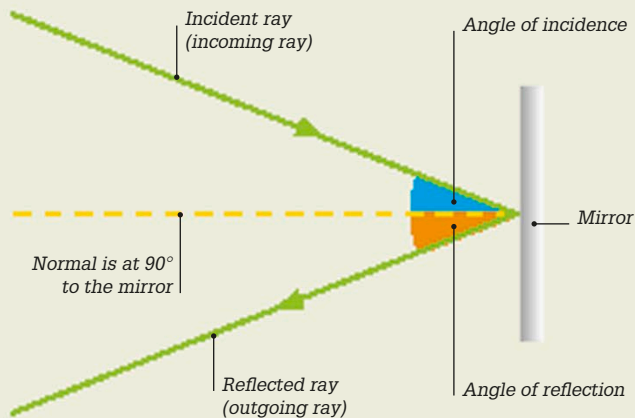


Reflection and refraction

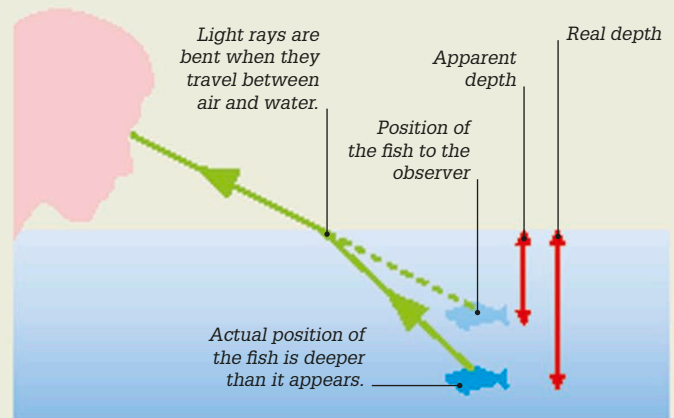
Light rays usually travel in straight lines. When the rays strike a flat, shiny surface, such as a mirror, they are reflected straight back to provide a clear, but reversed image. If the rays strike a rough surface, they bounce off in all directions and there is no clear reflected image. Light rays travel



Angles of incidence and reflection

The angle at which a light ray strikes a surface (angle of incidence) is equal to the angle at which the ray is reflected (angle of reflection). The angle is measured from an imaginary line called the normal.

through different materials, such as water, glass, or air, at different speeds. When a light ray passes from one material to another, its speed changes, causing the ray to change direction. This effect is known as refraction.

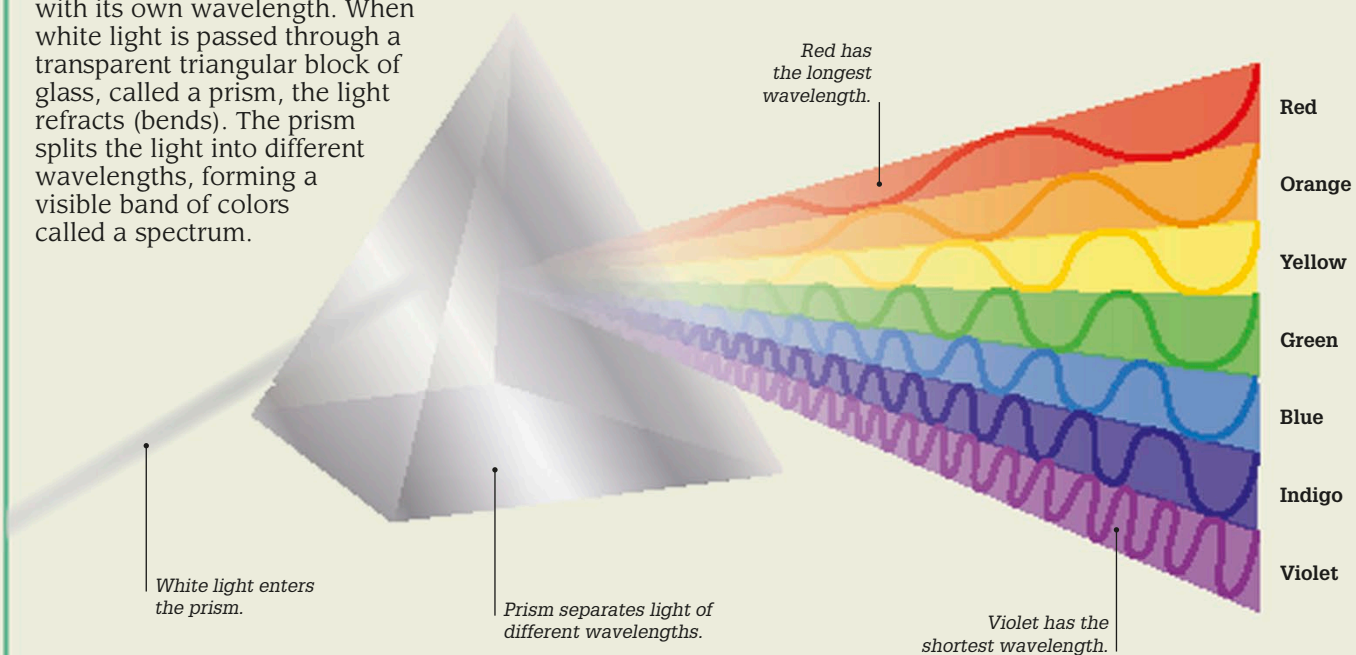


Real and apparent depth

When light rays pass from water to the lighter medium of air, they bend (refract). This means that if you look at an object in the water from an angle, it appears to be closer to the surface than it actually is.

Colored light spectrum

Light is made up of electromagnetic waves. "White light" is a mixture of many different colors of light, each with its own wavelength. When white light is passed through a transparent triangular block of glass, called a prism, the light refracts (bends). The prism splits the light into different wavelengths, forming a visible band of colors called a spectrum.



Splitting light

A prism bends light by different amounts according to its wavelength. There are seven main colors in the spectrum. Red has the longest wavelength and violet the shortest.